

AWS Landing Zone to AWS Control Tower - Migration Implementation Guide

Executive Summary

This document provides a comprehensive implementation framework for migrating from AWS Landing Zone (ALZ) solution to AWS Control Tower (CT). AWS Control Tower is the recommended managed service solution as AWS Landing Zone is currently in long-term support and will not receive any additional features. This migration will provide improved governance, reduced operational overhead, and access to AWS's ongoing feature enhancements.

Prerequisites

Before beginning the migration, ensure you have:

- New email addresses (never previously used with AWS accounts) for:
 - Log Archive account
 - Audit/Security account
- AWS SSO configured in the same region where you plan to deploy AWS Control Tower
- AWS Management account with no trusted access enabled for AWS Config or AWS CloudTrail
- Administrative access to your existing AWS Organizations

Migration Timeline

- **Estimated Total Duration:** 4-8 weeks (depends on environment complexity)
- **Planning Phase:** 1-2 weeks
- **Implementation Phase:** 3-6 weeks
- **Verification & Cleanup:** 1-2 weeks

Risk Assessment

Risk	Impact	Mitigation
Config recorder downtime	Temporary loss of compliance recording	Schedule during low-change periods; enroll accounts promptly after Config cleanup
SCP limit conflicts	Control Tower guardrails may fail to apply	Pre-consolidate SCPs; test thoroughly before rollout
Account enrollment failures	Delayed migration	Prepare rollback procedures; test with pilot accounts first
Service disruption	Business impact	Conduct thorough testing, establish maintenance windows, create detailed rollback plans

Phase 1: Deploy AWS Control Tower

Goal: Set up Control Tower in your AWS Organization.

Timeline: 1-2 days

Subtasks:

1. Check region compatibility

- Ensure AWS Control Tower and AWS SSO are in the same region
- Verify supported regions in AWS documentation
- Command to check SSO region: `aws sso-admin list-instances --region us-east-1`

2. Run pre-launch checks (automated)

- Navigate to AWS Control Tower console
- Select "Set up landing zone"
- Review and resolve any identified issues
- Validate prerequisites using AWS Control Tower readiness check
- Common issues: existing Config recorders, CloudTrail conflicts

3. Deploy Control Tower from the AWS Management Account

- Follow the AWS Control Tower Getting Started Guide
- Console path: AWS Management Console → Control Tower → "Set up landing zone"
- Select appropriate home region (must match SSO region)
- Configure organizational units structure
- Choose governance options
- Estimated deployment time: 60-90 minutes

Validation:

- Verify successful creation of AWS Control Tower landing zone
- Confirm automatic creation of Security/Audit and Log Archive accounts
- Check that foundational OUs (Security, Sandbox) are created
- Validate IAM Identity Center (SSO) configuration

Rollback Procedure:

- If deployment fails, review AWS Control Tower CloudWatch logs
- For critical failures, follow AWS Control Tower decommissioning procedure
- Command: `aws controltower delete-landing-zone`

Phase 2: Align Service Control Policies (SCPs)

Goal: Ensure Control Tower can apply its guardrails without exceeding SCP limits.

Timeline: 3-5 days

Subtasks:

1. Review existing SCPs applied via AWS Organizations

- List all current SCPs and their attachments
- Command: `aws organizations list-policies --filter SERVICE_CONTROL_POLICY`
- For each policy:

```
aws organizations list-targets-for-policy --policy-id <policy-id>
aws organizations describe-policy --policy-id <policy-id>
```

- Document policy content, affected OUs, and accounts

2. Consolidate or merge SCPs if you're near the limit (5 SCPs per OU/account)

- Identify policies that can be merged
- Create consolidated policy JSON documents
- Test policy effect with AWS Policy Simulator
- Implement consolidated policies using:

```
aws organizations create-policy --content file://consolidated-policy.json --name
"Consolidated-Policy" --type SERVICE_CONTROL_POLICY --description "Merged policies
for CT migration"
```

3. Ensure Control Tower roles are excluded from overly restrictive SCPs

- Modify SCPs to explicitly allow Control Tower service roles
- Add the following exemption to restrictive policies:

```

json
{
  "Effect": "Allow",
  "Action": "*",
  "Resource": "*",
  "Condition": {
    "StringLike": {
      "aws:PrincipalArn": [
        "arn:aws:iam::*:role/service-role/AWSControlTowerExecution",
        "arn:aws:iam::*:role/AWSControlTowerAdmin"
      ]
    }
  }
}

```

- Update policies using:

```

aws organizations update-policy --policy-id <policy-id> --content file://updated-policy.json

```

4. Test Control Tower permissions in existing accounts and OUs

- Create test cases for key Control Tower operations
- Validate that Control Tower service roles can perform required actions
- Use CloudTrail to verify successful API calls

Validation:

- Confirm that the SCP count per OU/account remains below the limit of 5
- Verify Control Tower roles have necessary permissions
- Test that Control Tower guardrails can be applied successfully

Rollback Procedure:

- Revert to original SCPs if issues are encountered
- Keep backup of original SCP definitions
- Use versioned policy JSON files for tracking changes

Phase 3: AWS Config Cleanup or Modification

Goal: Let Control Tower manage Config settings.

Timeline: 1-2 weeks (depends on number of accounts and regions)

Subtasks:

1. Option A: Delete Existing Config Setup (Recommended)

- From the management account, go to CloudFormation StackSets
- Delete stack instances of AWS-Landing-Zone-Baseline-EnableConfig

```
aws cloudformation delete-stack-instances \  
  --stack-set-name AWS-Landing-Zone-Baseline-EnableConfig \  
  --accounts <account-id-1> <account-id-2> ... \  
  --regions <region-1> <region-2> ... \  
  --no-retain-stacks
```

- Verify deletion using:

```
bash  
  
aws configservice describe-delivery-channels --region <region>  
aws configservice describe-delivery-channel-status --region <region>  
aws configservice describe-configuration-recorders --region <region>
```

- For each account and supported region, manually delete if needed:

```
bash  
  
aws configservice stop-configuration-recorder --configuration-recorder-name <name> --region <region>  
aws configservice delete-delivery-channel --delivery-channel-name <name> --region <region>  
aws configservice delete-configuration-recorder --configuration-recorder-name <name> --region <region>
```



- Check AWS Config aggregator authorizations:

```
aws configservice describe-aggregation-authorizations --region <region>  
aws configservice delete-aggregation-authorization --authorized-account-id <id> --  
authorized-aws-region <region> --region <region>
```

2. Option B: Modify Existing Config Setup

- Update templates:
 - aws-landing-zone-enable-config.template
 - aws-landing-zone-enable-config-role.template
- Modify according to AWS documentation for existing Config resources
- Update StackSets accordingly
- Run CLI commands for aggregation authorization
- Configure delivery channels to use Control Tower-compatible settings
- Update IAM roles to align with Control Tower requirements

Validation:

- Verify Config recorders and delivery channels are removed or properly modified

- Confirm no conflicts with Control Tower Config setup
- For Option A: Confirm complete removal in all affected accounts and regions
- For Option B: Validate Config settings align with Control Tower requirements

Rollback Procedure:

- For Option A: Re-deploy ALZ Config StackSets if needed
- For Option B: Revert template changes and update StackSets
- Document original Config settings for potential restoration

 **Warning:** Option A will cause a temporary gap in compliance recording. Schedule this activity during periods of minimal change and promptly enroll accounts in Control Tower.

Phase 4: Enroll Accounts & OUs

Goal: Bring existing accounts under Control Tower governance.

Timeline: 2-4 weeks (depends on number of accounts)

Subtasks:

1. Method 1: Register OU

- Register parent OU, then nested OUs
 - Console: Control Tower → Organization → Select OU → Register
 - Note: All accounts within the OU will be enrolled
- Ensure OU has ≤ 300 accounts
 - Split larger OUs if needed before registration
- Plan downtime/maintenance window
 - Communicate to stakeholders
 - Schedule outside business hours for critical accounts
 - Typical enrollment time: 30-60 minutes per account
- Be aware that new account creation is blocked during OU registration
 - Temporarily pause automated account creation processes

2. Method 2: Enroll Individual Accounts

- Use Account Factory via Service Catalog
 - Console: Service Catalog → Products → Account Factory
 - Select "Enroll Account"
- For custom AVM setups:
 - Retain old template for reference
 - Document customizations for potential re-implementation

- Create mapping between old and new customizations
- Provision new accounts using the Control Tower Account Factory product
 - Required parameters:
 - Account email
 - Account name
 - SSO user email
 - Organizational unit
 - IAM role name (default: AWSControlTowerExecution)
- Caveats:
 - Don't change Control Tower-managed Service Catalog products
 - Enroll one account at a time (no batch enrollment yet)
 - Consider automation scripts for large-scale enrollment

Validation:

- Verify accounts appear in Control Tower dashboard
- Confirm guardrails are applied correctly
- Check IAM Identity Center (SSO) access for enrolled accounts
- Validate AWS Config recorder and CloudTrail are functioning
- Test sample workflows to ensure operational continuity

Rollback Procedure:

- Document detailed steps for account de-registration
- Prepare procedures to restore original organization structure if needed
- Keep snapshots of organization configuration at key migration points

Phase 5: Migrate Core ALZ Accounts

Goal: Transition Log Archive, Audit/Security, and Shared Services accounts.

Timeline: 1-2 weeks

Subtasks:

1. Log Archive Account:

- Copy logs from ALZ S3 bucket to new Control Tower Log Archive bucket

```
bash
```

```
aws s3 sync s3://aws-landing-zone-logs-<id> s3://new-log-archive-bucket
```

- Migrate log analytics tools if needed
 - Update data source configurations
 - Test data access and queries
 - Validate log integrity and retention
- Do not upload logs into Control Tower-managed log bucket (restricted)
 - Create separate buckets for historical logs
 - Update retention policies
 - Configure appropriate permissions

2. **Security (Audit) Account:**

- Re-deploy services like:
 - GuardDuty

```
bash
```

```
aws guardduty create-detector --enable --finding-publishing-frequency FIFTEEN_MINUTE
```



- Security Hub

```
bash
```

```
aws securityhub enable-security-hub --enable-default-standards --region <region>
```

- SNS-based alerts

```
bash
```

```
aws sns create-topic --name SecurityAlerts --region <region>
```

- Configure topic policies and subscriptions
- Migrate any third-party integrations
 - Update API keys and endpoints
 - Test alert flows and incident response procedures
 - Document configuration changes

3. **Shared Services Account:**

- Retain existing account (CT doesn't create one)
- Enroll it into Control Tower using Account Factory
 - Follow Method 2 procedure from Phase 4
 - Carefully document and test account-specific configurations
- Migrate shared resources and update access policies
 - Test cross-account access patterns

- Validate service integrations

Validation:

- Verify log archive integrity and access
- Confirm security services are functioning correctly
- Test alert and notification flows
- Validate cross-account access for shared services
- Conduct full operational testing of migrated accounts

Rollback Procedure:

- Document detailed restore process for each account type
- Maintain access to original ALZ accounts until migration is fully validated
- Prepare emergency access procedures

Phase 6: Decommission AWS Landing Zone (ALZ)

Goal: Clean up ALZ resources safely.

Timeline: 1-2 weeks

Subtasks:

1. Pipeline & Provisioned Products:

- Disable CodePipeline transitions in ALZ pipelines

```
bash
```

```
aws codepipeline disable-stage-transition --pipeline-name <pipeline-name> --stage-name
```

- Terminate lz_core_* provisioned products if unused
 - Service Catalog → Provisioned products → Select product → Actions → Terminate
- Remove ALZ products from Service Catalog portfolios
 - Service Catalog → Portfolios → Select portfolio → Disassociate products
- Delete portfolios like:
 - "AWS Landing Zone – Baseline"
 - "AWS Landing Zone - Core"
 - "AWS Landing Zone - Add-On Products"
- Delete non-provisioned ALZ products (AWS-Landing-Zone-*)
- Delete CloudFormation AVM stacks and StackSets if unused

```
bash
```

```
aws cloudformation delete-stack --stack-name <stack-name>
```

2. S3 Buckets & Templates:

- Delete:
 - ALZ logging buckets (after migration)
 - ALZ config/artifacts buckets:
 - aws-landing-zone-configuration-*
 - landingzone-landingzonepipelineartifacts*

```
bash
```

```
aws s3 rb s3://<bucket-name> --force
```

- Delete initiation template stack (SO0044)

```
bash
```

```
aws cloudformation delete-stack --stack-name SO0044
```

3. StackSets to Clean (if not in use):

- AWS-Landing-Zone-Baseline-EnableCloudTrail
- AWS-Landing-Zone-Baseline-EnableConfig
- AWS-Landing-Zone-Baseline-EnableConfigRules
- AWS-Landing-Zone-Baseline-ConfigRole
- AWS-Landing-Zone-Baseline-EnableNotifications
- AWS-Landing-Zone-Baseline-PrimaryVPC
- AWS-Landing-Zone-Baseline-SecurityRoles
- IAM Password Policy, etc.

```
bash
```

```
aws cloudformation delete-stack-set --stack-set-name <stack-set-name>
```

4. Organization Cleanup:

- Delete ALZ-created OUs (not Control Tower OUs)

```
bash
```

```
aws organizations list-organizational-units-for-parent --parent-id <parent-id>
```

```
aws organizations delete-organizational-unit --organizational-unit-id <ou-id>
```

- Delete protect-cloudtrail-config SCP

```
bash
```

```
aws organizations delete-policy --policy-id <policy-id>
```

- Remove ALZ SSM Parameter Store values

```
bash
```

```
aws ssm delete-parameters --names "/aws/landing-zone/*"
```

- Delete KMS key alias:

```
bash
```

```
aws kms delete-alias --alias-name alias/AwsLandingZoneKMSKey
```

5. Terminate Old ALZ Accounts (if desired):

- Close ALZ Log Archive & Security accounts after migrating logs and services
- Follow AWS account closure process
- Retain accounts for 90 days before permanent closure if needed for audit purposes

Validation:

- Verify all ALZ resources are properly removed
- Confirm no lingering dependencies on ALZ components
- Check for orphaned resources or permissions
- Validate Control Tower operations after ALZ removal

Rollback Procedure:

- This phase represents a point of no return
- Ensure all validations are successful before proceeding
- Maintain backups of critical ALZ configuration artifacts

Appendix A: Command Reference

AWS Config Management Commands

bash

List configuration recorders

aws configservice describe-configuration-recorders --region <region>

Stop configuration recorder

aws configservice stop-configuration-recorder --configuration-recorder-name <name> --region <re

Delete configuration recorder

aws configservice delete-configuration-recorder --configuration-recorder-name <name> --region <

List delivery channels

aws configservice describe-delivery-channels --region <region>

Delete delivery channel

aws configservice delete-delivery-channel --delivery-channel-name <name> --region <region>

AWS Organizations Commands

bash

List organizational units

aws organizations list-organizational-units-for-parent --parent-id <parent-id>

List accounts in OU

aws organizations list-accounts-for-parent --parent-id <parent-id>

List policies

aws organizations list-policies --filter SERVICE_CONTROL_POLICY

Get policy details

aws organizations describe-policy --policy-id <policy-id>

Create policy

aws organizations create-policy --content file://policy.json --name <name> --type SERVICE_CONTR

Update policy

aws organizations update-policy --policy-id <policy-id> --content file://policy.json

Delete policy

aws organizations delete-policy --policy-id <policy-id>

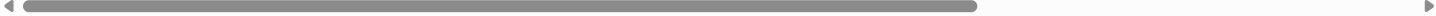
CloudFormation Commands

```
bash

# Delete stack
aws cloudformation delete-stack --stack-name <stack-name>

# Delete stack instances
aws cloudformation delete-stack-instances --stack-set-name <stack-set-name> --accounts <account>

# Delete stack set
aws cloudformation delete-stack-set --stack-set-name <stack-set-name>
```



Appendix B: Testing Procedures

Config Recorder Deletion Validation

```
bash

# Should return empty array if successful
aws configservice describe-configuration-records --region <region>
aws configservice describe-delivery-channels --region <region>
```

SCP Test Procedure

1. Create test IAM role with Control Tower service permissions
2. Attempt to assume role and perform common Control Tower operations
3. Verify successful API calls in CloudTrail

Account Enrollment Test Procedure

1. Enroll test account using Account Factory
2. Verify account appears in Control Tower dashboard
3. Confirm guardrails are applied
4. Test SSO access to account
5. Validate AWS Config and CloudTrail functionality

Appendix C: Common Issues and Resolutions

Control Tower Deployment Failures

- **Issue:** AWS SSO region mismatch
 - **Resolution:** Ensure AWS SSO is in the same region as Control Tower
- **Issue:** Existing Config recorder or delivery channel
 - **Resolution:** Delete using AWS CLI commands in Appendix A

- **Issue:** SCP limit exceeded
 - **Resolution:** Consolidate policies following Phase 2 guidelines

Account Enrollment Failures

- **Issue:** Missing AWSControlTowerExecution role
 - **Resolution:** Create role with proper trust relationship
- **Issue:** AWS Config conflicts
 - **Resolution:** Verify complete removal of existing Config resources
- **Issue:** SCP restrictions
 - **Resolution:** Modify SCPs to allow Control Tower service roles

Appendix D: Monitoring and Alerting Setup

- Create CloudWatch dashboards for migration progress
- Set up alarms for critical service health
- Configure SNS notifications for key migration events
- Establish rollback triggers based on service metrics

Appendix E: Post-Migration Activities

- Conduct comprehensive testing of governance controls
- Validate compliance reporting and monitoring
- Document new account provisioning procedures
- Train administrators on Control Tower operations
- Implement additional customizations as needed

References

Official AWS Documentation

1. [AWS Control Tower User Guide](#) - Comprehensive documentation on AWS Control Tower concepts and operations.
2. [Transitioning from AWS Landing Zone to AWS Control Tower](#) - AWS Prescriptive Guidance for the transition process.
3. [Walkthrough: Move from ALZ to AWS Control Tower](#) - Official migration walkthrough documentation.
4. [Enrolling Existing Accounts with AWS Control Tower](#) - Guidelines for enrolling existing accounts.
5. [AWS Organizations Guidance for AWS Control Tower](#) - Best practices for using AWS Organizations with Control Tower.

AWS Blog Posts and Whitepapers

6. [Migrate AWS Landing Zone solution to AWS Control Tower](#) - Official AWS Cloud Operations Blog post on the migration process.
7. [Migrating custom Landing Zone with RAM to AWS Control Tower](#) - Specific guidance for environments using AWS Resource Access Manager.
8. [Seamless Transition from an AWS Landing Zone to AWS Control Tower](#) - AWS Partner Network Blog on migration strategies.
9. [AWS Control Tower FAQs](#) - Frequently asked questions about AWS Control Tower.

Technical Resources

10. [AWS Config Developer Guide](#) - For managing AWS Config resources during migration.
11. [AWS Organizations User Guide](#) - For organizational structure and SCP management.
12. [AWS CloudFormation User Guide](#) - For managing CloudFormation stacks and StackSets.
13. [AWS CLI Command Reference](#) - For automating migration tasks.